

ABSOLUTE HODGE CYCLES AND THE PERIODS OF INTEGRALS OF ABELIAN VARIETIES

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ABSTRACT. We explain here how a suitable theory of “motives” recovers certain results of Shimura on the periods of abelian varieties of CM type [5]. It seemed instructive to begin with an informal description of the properties of motives and of their realizations; this is done in Section 1. Section 2 interprets certain results of Shimura by means of motives. Section 3 defines absolute Hodge cycles, and the corresponding theory of motives. Section 4 gives an upper bound for the transcendence degree of the field generated by the periods of an arbitrary abelian variety.

1. THE LANGUAGE OF MOTIVES

Let k be a field of characteristic 0. There are several definitions of “motives over k ” — it is not known whether they are equivalent — and each has its own virtues. In Section 3 we shall specify the one that is useful for us.

The motives over k form an additive category equipped with a tensor product. Every smooth projective variety X over k defines a motive $h(X)$ over k , the *motivic cohomology* of X , and every motive over k is obtained from a motive of this form (by cutting into pieces and Tate twisting). Finally, various cohomology theories for algebraic varieties correspond to realization functors on motives.

Every abelian variety A over k defines a motive $T(A)$ over k , its motivic H_1 , and the functor $A \mapsto T(A)$ is fully faithful. The advantage of embedding the category of abelian varieties into the category of motives is that one can then take tensor products. The resulting objects are, however, no longer geometric: there are no “functions” on a motive.

The *rank* of a motive is the dimension of any one of its realizations, which we now describe.

If $k = \mathbb{C}$, the *Betti realization* of a motive M is a free abelian group of finite rank $H_B(M)$ endowed with a *Hodge structure*, i.e. a decomposition

$$H_B(M) \otimes_{\mathbb{Z}} \mathbb{C} = \bigoplus_{p+q=n} H^{p,q} \quad (\overline{H}^{p,q} = H^{q,p}).$$

Let us illustrate this with the motive $T(A)$, namely the H_1 of an abelian variety A . One defines $H_B(T(A)) = H_1(A, \mathbb{Z})$, where $A = A(\mathbb{C})$ is regarded as a topological space.

Let $\text{Lie}(A)$ be the Lie algebra of the analytic group A , and let $J : H_1(A, \mathbb{Z}) \rightarrow \text{Lie}(A)$ be the homomorphism “integration along a 1-cycle.” There is an exact sequence of complex analytic groups

$$0 \longrightarrow H_1(A, \mathbb{Z}) \xrightarrow{J} \text{Lie}(A) \xrightarrow{\exp} A \longrightarrow 0.$$

The Hodge structure on $H_1(A, \mathbb{C})$ is such that $H_1(A, \mathbb{C}) = H^{0,-1} \oplus H^{-1,0}$. One obtains $H^{0,-1}$ as the kernel of the map $H_1(A, \mathbb{C}) \rightarrow \text{Lie}(A)$ obtained by extending $J : H_1(A, \mathbb{Z}) \rightarrow \text{Lie}(A)$ $\mathbb{C}$ -linearly.

Now let k be any field of characteristic 0 and fix an algebraic closure $\bar{k}$ of k . For each prime number ℓ , the ℓ -*adic realization* of a motive M is a free $\mathbb{Z}_\ell$ -module of finite type $H_\ell(M)$ (equipped moreover with an action of the Galois group of $\bar{k}$ over k). When $k = \mathbb{C}$, one has an isomorphism

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$H_\ell(M) \cong H_B(M) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$. If A is an abelian variety over k , and if $\bar{k}$ is an algebraic closure of k , the ℓ -adic realization of $T(A)$ is denoted $T_\ell(A)$; it is the $\mathbb{Z}_\ell$ -module

$$T_\ell(A) = \varprojlim_n A_{\ell^n}(\bar{k}) \quad (\text{where } A_{\ell^n}(\bar{k}) \text{ is the group of points of order } \ell^n \text{ in } A(\bar{k})).$$

If $k = \mathbb{C}$, the isomorphism $T_\ell(A) \cong H_B(T(A)) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$ is easy to describe. For every n , the projection of $\gamma \otimes x \in H_B(T(A)) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$ to $A_{\ell^n}(\mathbb{C})$ is computed as follows: if $m \in H_B(T(A))$ is such that $x - m \in \ell^n(H_B(T(A)) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)$, the exponential of the element $m/\ell^n \in H_1(A, \mathbb{Q})$ is the desired point of order ℓ^n .

The *de Rham realization* of M is a finite-dimensional vector space over k , denoted $H_{DR}(M)$, endowed with a decreasing filtration $F^\bullet$ (the *Hodge filtration*). When $k = \mathbb{C}$ there is an isomorphism $H_B(M) \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} H_{DR}(M)$. Under this isomorphism the Hodge filtration is such that $F^p = \bigoplus_{p' \geq p} H^{p', q}$. In the case of the motive $T(A)$, recall from [7] the existence of a “universal extension” E of A by a vector group, fitting into an exact sequence of algebraic groups

$$0 \longrightarrow H^1(A, \mathcal{O}_A)^* \longrightarrow E \longrightarrow A \longrightarrow 0.$$

One has $H_{DR}(T(A)) = \text{Lie}(E)$, and the Hodge filtration is given by

$$0 = F^1 \subset F^0 = H^1(A, \mathcal{O}_A)^* \subset F^{-1} = \text{Lie}(E).$$

If $k = \mathbb{C}$, one has $H_1(E, \mathbb{Z}) \xrightarrow{\sim} H_1(A, \mathbb{Z})$, hence a morphism

$$H_1(A, \mathbb{C}) \xleftarrow{\sim} H_1(E, \mathbb{C}) \xrightarrow{J} \text{Lie}(E) = H_{DR}(T(A)).$$

This is the desired isomorphism.

Examples. 1) Here $k = \mathbb{Q}$. We define the *Tate motive* $\mathbb{Z}(1)$; it is the H_1 of the algebraic group $\mathbb{G}_m$. To find its various realizations, one substitutes $\mathbb{G}_m$ for A in the definition of $T(A)$. One obtains:

- $H_B(\mathbb{Z}(1)) = 2\pi i \cdot \mathbb{Z}$, the kernel of $\exp : \mathbb{C} \rightarrow \mathbb{C}^*$;
- $H_B(\mathbb{Z}(1)) \otimes_{\mathbb{Z}} \mathbb{C} = H^{-1, -1}$;
- $H_\ell(\mathbb{Z}(1)) = \mathbb{Z}_\ell(1) = T_\ell(\mathbb{G}_m) = \varprojlim_n \mu_{\ell^n}(\bar{\mathbb{Q}})$;
- $H_{DR}(\mathbb{Z}(1)) = \mathbb{Q}$.

The isomorphism $H_B(\mathbb{Z}(1)) \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} H_{DR}(\mathbb{Z}(1)) \otimes_{\mathbb{Q}} \mathbb{C}$ sends $(2\pi i) \otimes 1$ to $2\pi i$. Recall the isomorphisms $H_1(\mathbb{G}_m) \cong H_1(\mathbb{P}^1 \setminus \{0, \infty\}) \cong H_2(\mathbb{P}^1)$. Moreover, for every smooth proper connected algebraic curve over $\mathbb{Q}$, $H_2(X)$ is isomorphic to $H_2(\mathbb{P}^1)$ (hence to $\mathbb{Z}(1)$). For every integer d , one defines a motive $\mathbb{Z}(d)$ so that $\mathbb{Z}(d) = \mathbb{Z}(d-1) \otimes_{\mathbb{Z}} \mathbb{Z}(1)$. One has $\mathbb{Z}(m+n) \cong \mathbb{Z}(m) \otimes_{\mathbb{Z}} \mathbb{Z}(n)$ for $m, n \in \mathbb{Z}$.

2) Let E be an elliptic curve over $\mathbb{Q}$. The motive $T(E)$ has rank 2. Hence the motive $\wedge^2 T(E)$ has rank 1, and one has $\wedge^2 T(E) \cong \wedge^2 H_1(E) \cong H_2(E) \cong \mathbb{Z}(1)$.

Let us make explicit the isomorphism $\wedge^2 T(E) \cong \mathbb{Z}(1)$ in the different realizations.

On $H_1(E(\mathbb{C}), \mathbb{R}) \cong \text{Lie}(E)$, the complex structure defines an orientation. If (e_1, e_2) is an oriented $\mathbb{Z}$ -basis of $H_B(T(E)) = H_1(E(\mathbb{C}), \mathbb{Z})$, then $e_1 \wedge e_2$ is a basis of $H_B(\wedge^2 T(E))$; it is sent to $2\pi i \in H_B(\mathbb{Z}(1))$.

The isomorphism $\wedge^2(T_\ell(E)) \cong \mathbb{Z}_\ell(1)$ is given by the alternating Weil pairing [6]. Finally, one has $\wedge^2 H_{DR}(T(E)) \cong \text{Hom}(H_{DR}^2(E), \mathbb{Q})$. Let ω and η be the usual differentials of the first and second kind. $H_{DR}^2(E)$ is generated by the cup-product $\omega \wedge \eta$ of their cohomology classes. Under the isomorphism $H_B(\wedge^2 T(E)) \otimes_{\mathbb{Z}} \mathbb{C} \rightarrow H_{DR}(\wedge^2 T(E)) \otimes_{\mathbb{Q}} \mathbb{C}$, the element $e_1 \wedge e_2$ maps to the homomorphism that sends $\omega \wedge \eta \in H_{DR}^2(E)$ to

$$\begin{vmatrix} \omega_1 & \omega_2 \\ \eta_1 & \eta_2 \end{vmatrix} \in \mathbb{C}.$$

Compatibility between the Betti and de Rham realizations of the isomorphism $\Lambda^2(T(E)) \cong \mathbb{Z}(1)$ is equivalent to the *Legendre relation*

$$\omega_1 \eta_2 - \omega_2 \eta_1 = 2\pi i.$$

3) Let A be an abelian variety over $\mathbb{Q}$. To the cohomology class of a hyperplane section of A in a projective embedding there corresponds a morphism of motives $\Lambda^2 T(A) \rightarrow \mathbb{Z}(1)$. In Betti cohomology this corresponds to an alternating form $2\pi i \cdot E$ on $H_B(T(A))$, with values in $2\pi i \cdot \mathbb{Z}$. In de Rham cohomology one likewise has an alternating form E_{DR} on $H_{DR}(T(A))$ with values in $\mathbb{Q}$. It is classical that F^0 is a totally isotropic subspace. It follows that one can find differentials of the first kind $\omega_1, \dots, \omega_g$ on A and differentials of the second kind $\eta_1, \dots, \eta_g$ on A such that

$$E_{DR}(x, y) = \sum_{i=1}^g ((\omega_i \cdot x)(\eta_i \cdot y) - (\omega_i \cdot y)(\eta_i \cdot x)).$$

Compatibility between E and E_{DR} is expressed as follows. Let $(e_1, \dots, e_{2g})$ be a basis of $H_B(T(A))$, let Ω be the $g \times 2g$ matrix with entries $\Omega_{ij} = \int_{e_j} \omega_i$, and let N be the $g \times 2g$ matrix with entries $N_{ij} = \int_{e_j} \eta_i$. Then

$${}^t N \cdot \Omega - \Omega \cdot N = 2\pi i \cdot E.$$

Finally, in the ℓ -adic realization one finds an alternating form on $T_\ell(A)$ with values in $\mathbb{Z}_\ell(1)$: it is the Weil pairing attached to the divisor of the hyperplane section [6].

2. MOTIVES ATTACHED TO CM ABELIAN VARIETIES

Let k be an algebraically closed subfield of $\mathbb{C}$ — for example $\bar{\mathbb{Q}}$ — and let $\mathcal{A}$ be the full subcategory of the category of motives over k generated by the $T(A)$ of abelian varieties, under tensor product, duals, and direct summands. The functor “Betti realization of the motive obtained from M/k after extension of scalars to $\mathbb{C}$ ” sends $\mathcal{A}$ to the category of Hodge structures. It follows formally from Theorem 1 of Section 3 that this functor is fully faithful.

Let E be a CM field, let S be the set of its complex embeddings, and let $- : S \rightarrow S$ denote complex conjugation. To each function $\varphi : S \rightarrow \mathbb{Z}$ satisfying

$$(*) \quad \varphi(\sigma) + \varphi(\bar{\sigma}) \text{ is independent of } \sigma \in S,$$

attach a Hodge structure H_φ endowed with a module structure over the ring of integers $\mathcal{O}_E$ of E : $H_\varphi = \mathcal{O}_E$, with its evident module structure, and in the decomposition $H_\varphi \otimes_{\mathbb{Z}} \mathbb{C} = E \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C}^S$, the factor $\mathbb{C}$ indexed by $\sigma \in S$ is assigned Hodge type $(\varphi(\sigma), \varphi(\bar{\sigma}))$.

One checks that the H_φ belong to the image of $\mathcal{A}$, indeed already to the image of the subcategory generated by CM abelian varieties. This allows one to define the motive M_φ over $\bar{\mathbb{Q}}$ as the motive whose Betti realization is H_φ .

The de Rham realization $H_{DR}(M_\varphi)$ is a vector space over $\bar{\mathbb{Q}}$ endowed with an action of E . For $\sigma \in S$, let $H_{DR}(M_\varphi)_\sigma$ be the $\bar{\mathbb{Q}}$ -subspace on which $x \in E$ acts as $\sigma(x) \in \bar{\mathbb{Q}}$. Each of these spaces is one-dimensional, generated by an element ω_σ , and $H_{DR}(M_\varphi)$ is their direct sum. Via the isomorphism $H_B(M_\varphi) \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} H_{DR}(M_\varphi) \otimes_{\bar{\mathbb{Q}}} \mathbb{C}$, an element e of $E = H_B(M_\varphi) \otimes_{\mathbb{Q}} \mathbb{Q}$ corresponds to a family (e_σ) with $e_\sigma \in H_{DR}(M_\varphi)_\sigma$. One checks immediately that the complex number $p(\sigma, \varphi)$ defined by

$$p(\sigma, \varphi) e_\sigma = \omega_\sigma$$

is well defined up to multiplication by an element of $\bar{\mathbb{Q}}^*$, independently of both e and the choice of ω_σ . This justifies the notation.

The relation between these periods and Shimura’s invariants $p_E(\sigma, \varphi)$ (see [5]) is

$$p_E(\sigma, \varphi) = \frac{p(\sigma, \varphi)}{(2\pi i)^{\varphi(\sigma)}} \pmod{\bar{\mathbb{Q}}^*},$$

which is simply a reformulation of the definitions.

The identities between Shimura's periods correspond to algebraic properties of the motives M_φ . Before summarizing them in a table, let us introduce some notation for an extension F/E of CM fields. Write S_F (resp. S_E) for the set of complex embeddings of F (resp. E). For $\sigma \in S_F$, write $\sigma|_E \in S_E$ for the restriction of σ to E .

For $\varphi : S_E \rightarrow \mathbb{Z}$ satisfying (*), let $\text{Inf}(\varphi)$ be the function on S_F defined by $\text{Inf}(\varphi)(\sigma) = \varphi(\sigma|_E)$. This function again satisfies (*). For $\varphi : S_F \rightarrow \mathbb{Z}$ satisfying (*), let $\text{Res}(\varphi)$ be the function on S_E defined by $\text{Res}(\varphi)(\sigma) = \sum_{\tau|_E=\sigma} \varphi(\tau)$; it too satisfies (*).

Identities among periods	Relations among motives
1. $p(\sigma, \varphi' + \varphi'') = p(\sigma, \varphi') p(\sigma, \varphi'')$	1. $M_{\varphi' + \varphi''} \cong M_{\varphi'} \otimes_E M_{\varphi''}$
2. For a CM extension F/E , $\sigma \in S_F$, and $\varphi : S_E \rightarrow \mathbb{Z}$ satisfying (*); $p(\sigma, \text{Inf}(\varphi)) = p(\sigma _E, \varphi)$.	2. $M_{\text{Inf}(\varphi)} \cong M_\varphi \otimes_E F$
3. For $\sigma \in S_E$ and $\varphi : S_F \rightarrow \mathbb{Z}$ satisfying (*); $p(\sigma, \text{Res}(\varphi)) = \prod_{\tau _E=\sigma} p(\tau, \varphi)$.	3. M_φ is an F -motive of rank 1, hence an E -motive of rank $d = [F : E]$, and $M_{\text{Res}(\varphi)} \cong \bigwedge_E^d M_\varphi$.
4. $p(\sigma, \mathbb{1}) = 2\pi i$	4. $M_{\mathbb{1}} = \mathbb{Z}(-1)$, the dual of the Tate motive.

These identities among periods recover Theorems 1–4 of Shimura's article [5]. Note that in [5] Shimura relates the periods $p_E(\sigma, \varphi)$ to the values at a special point of Hilbert modular forms rational over $\bar{\mathbb{Q}}$.

3. ABSOLUTE HODGE CYCLES AND MOTIVES

Let X be a smooth projective variety over $\mathbb{C}$. A *Hodge cycle* of codimension d on X is an element of $H^{2d}(X, \mathbb{Q})$ of type (d, d) , or equivalently an element of

$$H^{2d}(X, \mathbb{Q})(d) \stackrel{\text{def}}{=} H^{2d}(X, \mathbb{Q}) \otimes_{\mathbb{Z}} \mathbb{Z}(d)$$

of type $(0, 0)$. Note that for $x \in H^{2d}(X, \mathbb{Q})(d)$ it is equivalent to say that x is of type $(0, 0)$ or that x lies in F^0 . A cohomology class coming from an algebraic subvariety of X is a Hodge cycle. Hodge conjectured that every Hodge cycle should be a linear combination of cohomology classes of subvarieties. Since that conjecture seems inaccessible, one introduces a notion that “arithmetizes” the notion of Hodge cycle.

The cycles we define will possess all the visible attributes of algebraic cycles, namely realizations in ℓ -adic and de Rham cohomology. An *absolute Hodge cycle* of codimension d on a smooth projective variety X over a field k of characteristic 0 is a collection of cohomology classes in the groups $H_{\text{et}}^{2d}(X_{\bar{k}}, \mathbb{Q}_\ell)(d)$ (resp. $H_{DR}^{2d}(X)(d)$) satisfying the following two conditions:

- the element of $H_{DR}^{2d}(X)(d)$ lies in F^0 ;
- for every embedding $k \hookrightarrow \mathbb{C}$, the various cohomology classes correspond (via the isomorphisms of Section 1) to a rational class in the Betti cohomology of $X \otimes_k \mathbb{C}$.

Thus one has inclusions

$$(\text{algebraic cycles}) \subset (\text{absolute Hodge cycles}) \subset (\text{Hodge cycles}).$$

Using this definition, let us now specify the category of motives used in Sections 1 and 2. Let $\mathcal{V}(k)$ first denote the category of smooth projective varieties over k . One constructs an *additive* category $\mathcal{M}(k)$ in which the groups $\text{Hom}(M, N)$ are $\mathbb{Q}$ -vector spaces, and which is equipped with the following data:

- a) a tensor product $\otimes$, associative, commutative, and distributive with respect to direct sums of objects;
- b) a contravariant functor H^* from $\mathcal{V}(k)$ to $\mathcal{M}(k)$, bijective on objects, compatible with sums, and taking products to tensor products.

For connected X and Y one has in $\mathcal{M}(k)$

$$\mathrm{Hom}(H^*(X), H^*(Y)) = Z_{h.a.}^{\dim(Y)}(X \times Y),$$

where $Z_{h.a.}^d(Z)$ denotes the $\mathbb{Q}$ -vector space of absolute Hodge cycles of codimension d on Z . The category of motives $\mathcal{M}(k)$ is then constructed exactly as in [2], with algebraic cycles replaced by absolute Hodge cycles. Thanks to this new definition, the Künneth components of the diagonal of $X \times X$ are absolute Hodge cycles, and hence one has in $\mathcal{M}(k)$ a decomposition

$$H^*(X) = \bigoplus_i H^i(X).$$

The following result was presented at the I.H.É.S. in 1979.

Theorem 1. *Every Hodge cycle on an abelian variety is absolute Hodge.*

We also note the following.

Corollary 1 (Borovoi[✓]). *Let A be an abelian variety over a field k of characteristic 0, and let ξ be a Hodge cycle on A . Then the ℓ -adic realizations ξ_ℓ of ξ are fixed by an open subgroup of $\mathrm{Gal}(\bar{k}/k)$.*

4. PERIODS OF INTEGRALS OF ABELIAN VARIETIES

Let A be a complex abelian variety, and let

$$H_1(A, \mathbb{C}) = H^{0,-1} \oplus H^{-1,0}$$

be the Hodge decomposition of $H_B(A)$ (cf. Section 1). Regard $\mathrm{GL}(H_1(A))$ as an algebraic group over $\mathbb{Q}$. There is then a morphism of complex algebraic groups

$$\mu : \mathbb{G}_m \rightarrow \mathrm{GL}(H_1(A, \mathbb{C}))$$

such that $\mu(z)$ acts by the homothety of ratio z on $H^{-1,0}$ and by the identity on $H^{0,-1}$. The *Hodge group* G of A is the Zariski closure of the subgroup $\mu(\mathbb{C}^*)$ of $\mathrm{GL}(H_1(A))$. In other words, G is an algebraic subgroup over $\mathbb{Q}$ of $\mathrm{GL}(H_1(A))$. A $\mathbb{Q}$ -rational regular function on a Zariski-open subset of G vanishes on G if and only if it vanishes on $\mu(\mathbb{C}^*)$. One checks easily that G contains the scalar homotheties of $H_1(A)$. A second description of G , very useful in practice, is obtained by considering all rational tensors of type $(-N/2, -N/2)$ in the tensor powers $\bigotimes^N H_1(A)$. Then G is the subgroup of $\mathrm{GL}(H_1(A))$ consisting of those linear transformations g for which there exists a scalar μ such that

$$g \cdot t = \mu^m t$$

for every tensor t of type (m, m) . Finally, one shows that G is the group fixing all absolute Hodge cycles of type $(0, 0)$ in the mixed tensor spaces built from $H_1(A)$.

Now assume that A is defined over a subfield k of $\mathbb{C}$. Put $g = \dim(A)$.

Theorem 2. *Let $\omega_1, \dots, \omega_g$ be differentials of the first kind on A , and let $\eta_1, \dots, \eta_g$ be differentials of the second kind, such that*

$$(\omega_1, \dots, \omega_g, \eta_1, \dots, \eta_g)$$

is a basis of $H_{DR}^1(A)$. Let K be the subfield of $\mathbb{C}$ generated by k and by the periods

$$\int_\gamma \omega_i, \quad \int_\gamma \eta_i \quad (1 \leq i \leq g, \gamma \in H_1(A, \mathbb{Z})).$$

Then the transcendence degree of K over k is bounded above by the dimension of G .

Proof. K is the field of definition of the isomorphism

$$H_{1,B}(A) \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} H_{1,DR}(A) \otimes_k \mathbb{C}$$

given by integration along cycles. This isomorphism preserves the cohomology classes of absolute Hodge cycles. Let P be the algebraic variety of isomorphisms with this property. By the definition of G , together with Theorem 1, P is a principal homogeneous space under G , defined over k , so $\dim(P) = \dim(G)$. One concludes by the following lemma.

Lemma 1. *Let z be a point of $\mathbb{A}^N(\mathbb{C})$. The transcendence degree of $k(z)$ over k is equal to the dimension of the Zariski closure of $\{z\}$ over k .*

This is tautological. □

Remark 1. From Example 3 of Section 1 it follows that $2\pi i \in K$. One can moreover show that the transcendence degree of K over $k(2\pi i)$ is at most $\dim(G) - 1$.

Corollary 2. *With the notation of Theorem 2, assume moreover that A is simple of CM type. Then the transcendence degree of K over k is at most $g + 1$.*

Proof. It suffices to show that $\dim(G) \leq g + 1$. Now $\mu(\mathbb{C}^*)$ commutes with the action of the field E on $H_1(A, \mathbb{Q})$. It follows that G also commutes with this action, and therefore G is an algebraic torus. Moreover, G preserves, up to scalar, the alternating form associated with a hyperplane section (see Example 3 of Section 1). Hence the claim. □

Note that in this last situation one also has the lower bound

$$\dim(G) \geq 2 + \log_2(g),$$

due to Ribet [3].

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